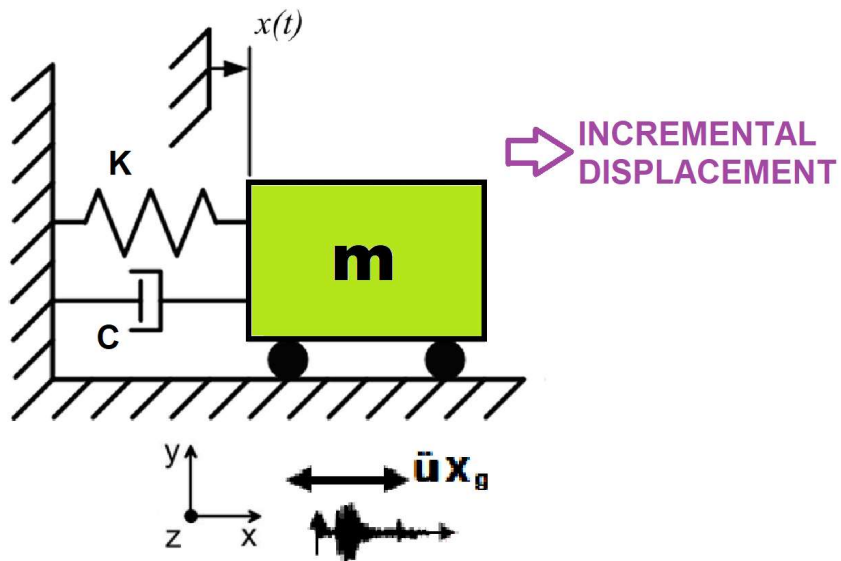


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL YIELD STRENGTH FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

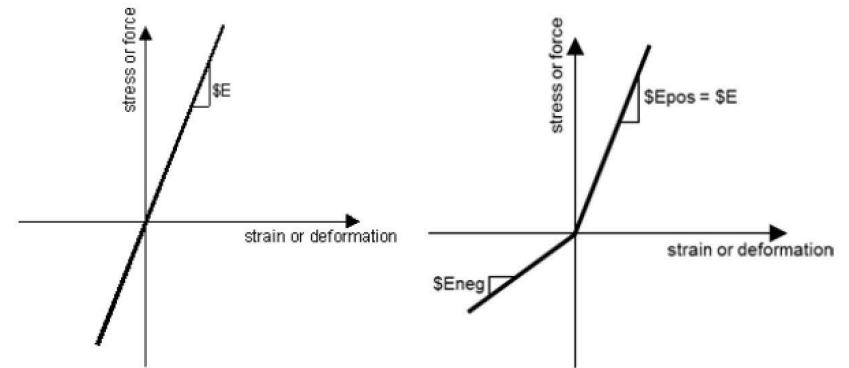
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

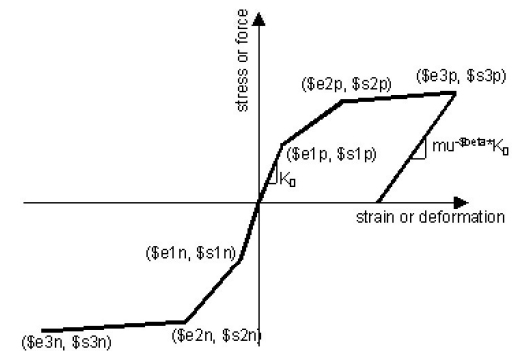
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

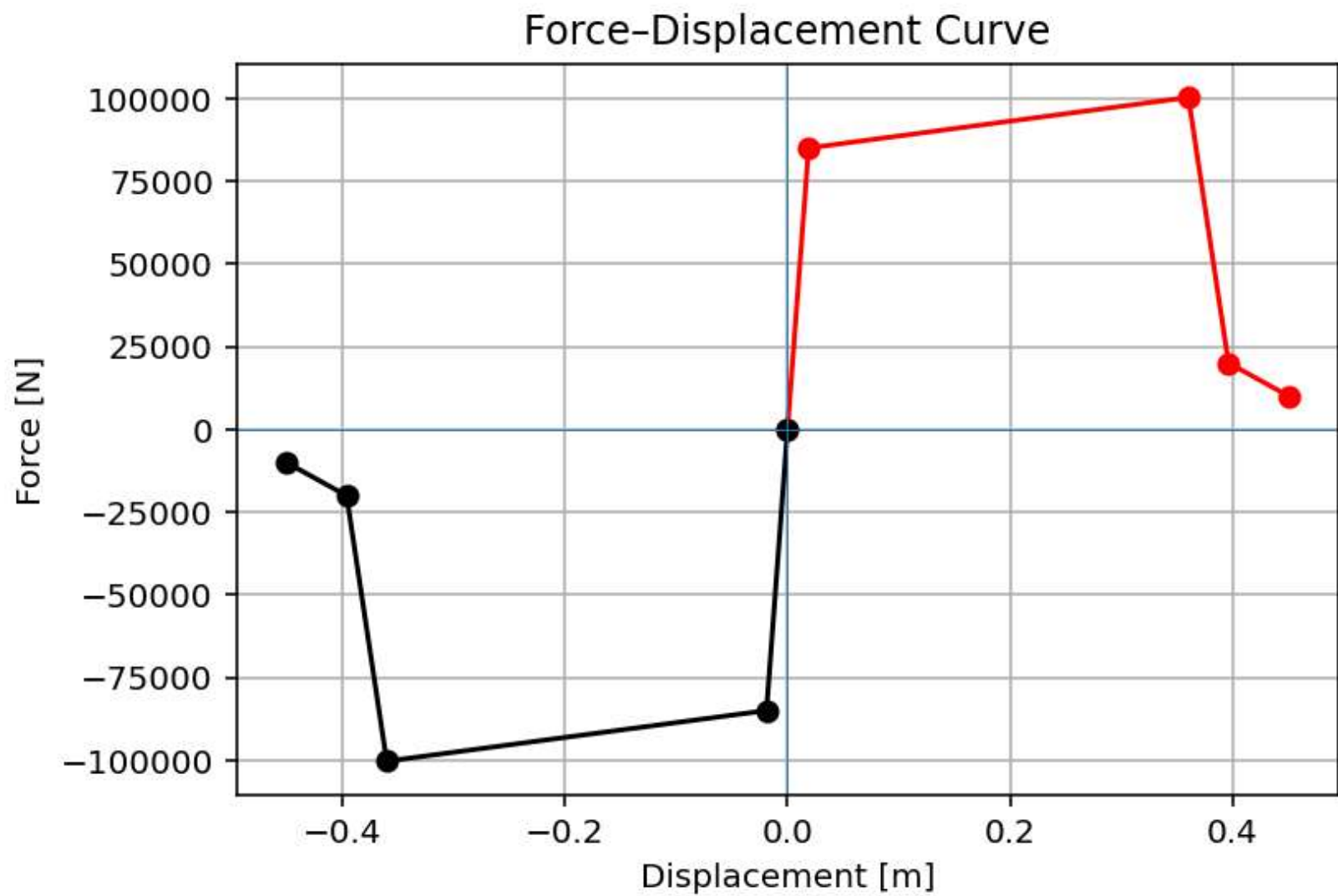
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



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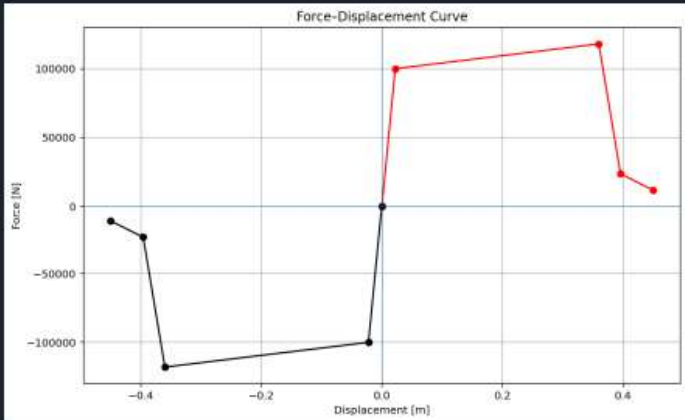
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TIMIZATION_DECI_SDOF\FY_OPTIMIZATION_DECI_SDOF.py

FY_OPTIMIZATION_DECI_SDOF.py x DISSIPATED_ENERGY_WITH_PLOT_FUN.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF OPTIMUM YIELD STRENGTH FROM ENERGY DISSIPATION CAPACITY INDEX U
4 # FINITE-DIFFERENCE NEWTON ITERATION
5 #
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPEN
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
8 #
9 # P(t) = P0 sin(wt)
10 # P(t) = P0 exp(-0.05wt) sin(wt)
11 #
12 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EV
13 # OF ENERGY DISSIPATION CAPACITY INDEX
14 #
15 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
16 # EMAIL: salar.d.ghashghaei@gmail.com
17 #####
18 """
19 Nonlinear Seismic Performance Assessment of a SDOF:
20 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and E
21
22 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
23 for performance-based earthquake engineering. The 2D model incorporates both material nonl
24 (elastic-perfectly plastic or hysteretic steel with strain hardening)
25 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
26 and large displacements-essential for collapse assessment.
27
28 Eight analysis protocols are implemented:
29 (1) [PERIOD] : Structural Period
30 (2) [STATIC] : Gravity load analysis establishing dead load state
31 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
32 and plastic mechanism identification
33 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
34 pinching behavior, and energy dissipation degradation
```

34 %

Force-Displacement Curve



Displacement [m]	Force [N]	Path Type
-0.4	-100000	Loading
-0.2	-50000	Loading
0.0	0	Loading
0.2	100000	Loading
0.4	100000	Loading
0.4	25000	Unloading
0.2	0	Unloading
0.0	0	Unloading
-0.2	-50000	Unloading
-0.4	-100000	Unloading

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OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Spyder (Python 3.12)

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C:\Users\ DELL\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...TIMIZATION_DEC1_SDOF\FY_OPTIMIZATION_DEC1_SDOF.py

FY_OPTIMIZATION_DEC1_SDOF.py x DISSIPATED_ENERGY_WITH_PLOT_FUN.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF OPTIMUM YIELD STRENGTH FROM ENERGY DISSIPATION CAPACITY INDEX I
4 # FINITE-DIFFERENCE NEWTON ITERATION
5 #
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPEN
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
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```

Console 1/A x

```
+-----+
| lambda | omega | period | frequency |
+-----+
| 9.000e+02 | 30.0000 | 0.2094 | 4.7746 |
+-----+

Time: 20.0000, Displacement: 0.0120 mm, Reaction: -20.00 N

SEISMIC ANALYSIS DONE.

Xmax: 166719.0593285352
Dissipated Energy from Earthquake= 32412.64 N·mm
Dissipated Energy from Cyclic Displacement= 162063.22 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -3.704793840597631e-07
DF: -0.0003704793556380537
DX: 2.876851536854367e-11
IT: 7 - RESIDUAL: 2.876851536854367e-11 X: 166719.0583285352

Optimum X: 166719.0583
Iteration Counts: 7
Convergence Residual: 2.8768515369e-11

Total time (s): 60.9844
```

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Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 73, Col 1 UTF-8 CRLF RW Mem 38%

Dissipated Energy from Earthquake= 32412.64 N·m

Dissipated Energy from Cyclic Displacement= 162063.22 N·m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -3.704793840597631e-07

DF: -0.0003704793556380537

DX: 2.876851536854367e-11

IT: 7 - RESIDUAL: 2.876851536854367e-11 X:

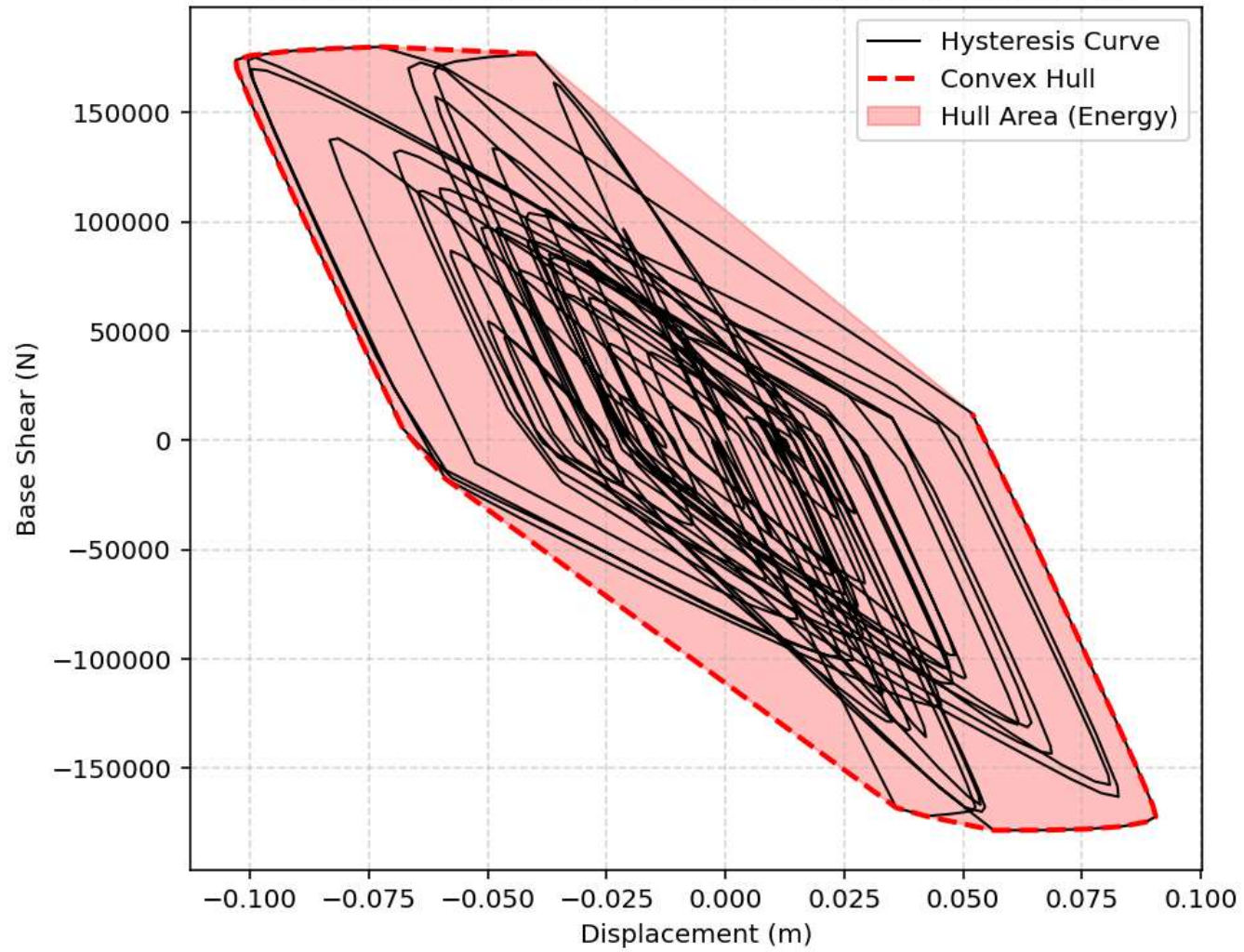
166719.0583285352

Optimum X: 166719.0583

Iteration Counts: 7

Convergence Residual: 2.8768515369e-11

Earthquake Response - Dissipated Energy (Convex Hull)



Cyclic Loading - Dissipated Energy (Convex Hull)

